

To do: Make a submission

Opened: Thursday, 9 April 2026, 12:05 AM

Due: Thursday, 16 April 2026, 11:55 PM

Assignment instructions

For this assignment you will apply what you have learned in this unit about database systems and answer the following questions.

1.
 - a) Design an entity-relationship data model using DIA software for a car-insurance company whose customers own one or more cars each. There are zero to any number of accidents recorded for each car.
 - (b) Provide an example of each of the following three concepts within your model:
 - Primary key
 - Composite attribute
 - Multi-valued attribute
 - c) Follow these [instructions for downloading and using Dia Diagramming Editor](#) to create your entity relationship diagram.
2.
 - a) Using the concepts of relational databases, develop a set of tables with several entries in each table from the model designed in question 1.(a).
 - b) Using these tables, provide an example of each of the following:
 - A left join
 - A right join
 - A full outer-join
3.
 - (a) Give examples for each of the following four mapping cardinalities:
 - one-to-one
 - many-to-one
 - one-to-many
 - many-to-many
 - b) Explain why the cardinality is appropriate for the relationship in each example.
 - one-to-one
 - many-to-one
 - one-to-many
 - many-to-many
 - many-to-many

Submission instructions

- Write the questions before the answer.
- Edit for spelling and grammar errors.
- Use APA citations and references if you use ideas from the readings or other sources. For assistance with APA formatting, view the [Learning Resource Center: Academic Writing](#).
- Make sure you submit both files i.e., one-word file (containing ER diagram along with its explanation) and the other with DIA as an attachment in your assignment.

Your assignment will be assessed by your instructor using the rubric below



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Submission status

Attempt number	This is attempt 1.
Submission status	No submissions have been made yet
Grading status	Not graded
Time remaining	1 hour 5 mins remaining
Last modified	-
Submission comments	▶ Comments (0)

Grading criteria

Q1 (a)	The ER model effectively represents the problem description. 20 points	The ER model is mostly correct (with only minor error(s)). 16 points	An ER model is created that is partially correct. 10 points	An ER Model is created, but there are significant errors in the model. 5 points	No ER model has been provided. 0 points
Q1 (b)	All the three concepts are accurately identified. 20 points	Two of the concepts are correctly identified. 16 points	One of the concepts is accurately identified. 10 points	None of the concepts has been defined accurately. 0 points	
Q2 (a)	Tables are provided with all effective examples. 15 points	Tables are provided with mostly correct examples. 12 points	Tables are provided with least effective examples. 5 points	Tables are not provided. 0 points	
Q2 (b)	All the three joins are correct. 15 points	Any two of the joins are correct. 12 points	Only one of the joins is correct. 8 points	Joins are not executed correctly. 6 points	No response for Q2b. 0 points
Q3 (a)	All four examples are correct. 15 points	Three of the examples are correct. 12 points	Two of the examples are correct. 8 points	None of the examples is correct. 6 points	
Q3 (b)	Explanations are clear, insightful, and details are provided. 15 points	Explanations show general understanding of the concept. 10 points	Explanations do not fully demonstrate understanding. 8 points	Explanations are incomplete or incorrect. 3 points	No explanation is present. 0 points